HETEROGENEITY UNDER OBSERVABILITY

An Acquisition-Cost Framework for Assessing Diversity as a Cyber-Physical Defense

Response to DARPA-SN-27-17, Formal Learning and Abstractions for System Heterogeneity (FLASH) · Submitted by AionSystem · Technical POC: Sheldon K. Salmon · AionSystem@outlook.com · (315) 460-6977 · UNCLASSIFIED · November 2026

SUMMARY

FLASH proposes to generate many provably correct, meaningfully different implementations of a specification and to use the diversity of failure modes and attack surfaces to reduce the durability and transferability of adversary knowledge. RFI 1(iv) asks for key assumptions, limitations, and evaluation methods. This response names the adversary's acquisition model as the assumption that decides heterogeneity's value, gives a method for determining which acquisition class applies to a program (§2.1), proposes a metric that predicts the adversary's burden under each class (§6), and gives an evaluation protocol that tests the prediction (§8).

Three adversary classes are separated by how the adversary acquires knowledge. A class-attacking adversary (A_1) studies the design family from public information and never accesses the deployment. A capture-then-exploit adversary (A_2-hybrid) captures one deployed instance, studies it, and deploys exploits across the fleet. A real-time adversary (A_2-pure) probes the deployed instance without study or persistent access. Heterogeneity raises the burden under A_1 and A_2-hybrid by the same structural multiplier, $1 + H_{d,variant} \cdot (N-1)$ cycles of per-variant work, where $H_{d,variant}$ is the fraction of variants whose cheapest exploit path is independent. For static or slow-dynamic heterogeneity it does not change the burden under A_2-pure. Whether the absolute value is larger under A_2-hybrid depends on whether $c_{capture} + c_{exploit}$ exceeds c_{public}.

A program that builds heterogeneity without determining which class dominates cannot attribute resilience to heterogeneity. The response also covers invariants and degrees of freedom (§5), independence of co-generated assurance and explanation (§9), and reusable libraries (§10). The anchor class is unmanned surface and undersea vehicles (USV/UUV). Evidence for the central claim is the §8 protocol, which has not been run. RFI coverage is mapped in Annex A.

1. DOMAIN AND APPLICATION CLASS [RFI 1(i)]

Domain. Autonomous systems operating in contested environments, where an adaptive adversary can observe, interact with, and attempt to compromise the deployed system. The unit of analysis is a *design family*: a set of functionally equivalent implementations generated from one specification, deployed so that each member is a realization, not a type. Heterogeneity is a property of the family, because a single artifact cannot be heterogeneous with respect to itself.

Anchor. Unmanned surface vessels (USVs) and unmanned undersea vehicles (UUVs) performing autonomous maritime patrol: persistent presence in a contested maritime area, sensor-fusion contact classification, and collision avoidance under COLREGs. Platforms are mission-persistent, operate without continuous operator control, use intermittent encrypted command-and-control links, and work in waters within physical reach of adversary vessels and

aircraft. They are forward-deployed by physical necessity. Hull form, propulsion, sensor suite, communications, and autonomy stack vary across vendors and platform classes; whether that variation moves critical failure nodes is the question the framework is built to answer. The anchor is named from public regulatory documents (COLREGs; 33 CFR Part 83) and general industry practice (L3).

Three conditions for heterogeneity to pay. (a) *Genuine degrees of freedom:* the family varies in ways that provably preserve required behavior (§5). (b) *Acquisition model:* the adversary must acquire knowledge of each variant, which excludes full state-observability through real-time probing (§2). (c) *Verification economy:* per-variant verification cost does not exceed the value of the diversity (§8, criterion 6). Condition (b) decides whether heterogeneity pays.

2. THREAT ENVIRONMENT AND ADVERSARY CLASSES [RFI 1(ii), 1(iv)]

Rationale. Contested autonomous systems concentrate national-security value because they place decision and effect at the edge, with hardware that degrades and software that is updated. The threat is an adversary who can choose when and how to specialize. Capture of unmanned maritime vehicles by adversary forces is documented: on 15 December 2016 a Chinese Navy vessel seized a U.S. Navy underwater glider about 50 nautical miles northwest of Subic Bay and returned it on 20 December [R1, R2]. A deployed maritime vehicle can come into adversary hands. Cyber-intrusion and supply-chain paths to the same artifacts are per-platform questions addressed by §2.1.

	A_1 class-attacking	**A_2-hybrid capture-then-exploit**	**A_2-pure real-time**
Acquires	Design-family knowledge from public or purchased information	A deployed instance (physical capture or cyber intrusion), with time to study	Transient interaction with the deployed instance; no capture, no study
Burden, homogeneous family	$1 \times c_{public}$	$c_{capture} + c_{exploit}$ (fleet compromised)	1 × per-engagement cost
Burden, $H_{d,variant} > 0$	$[1 + H_{d,variant} \cdot (N-1)] \times c_{public}$	$[1 + H_{d,variant} \cdot (N-1)] \times (c_{capture} + c_{exploit})$	1 × per-engagement cost
Effect of heterogeneity	Same multiplier as A_2-hybrid, lower unit cost	Same multiplier as A_1; higher unit cost if $c_{capture} + c_{exploit}$ exceeds c_{public}	None, for static or slow-dynamic heterogeneity

The classes are a taxonomy, not an exclusive typing: an adversary can pair A_1 preparation with A_2-pure engagement. The burden calculation uses the class that dominates the adversary's work.

When the A_2-hybrid formula holds. (i) *No missed transfer:* the §6 test has identified every transfer between variants. (ii) *Post-test $H_{d,variant}$:* the value is the output of the full §6 procedure. (iii) *Cluster structure:* the non-independent $(1 - H_{d,variant})$ variants form one transfer cluster; with $k_{clusters}$ transfer-equivalence classes the cycle count is $H_{d,variant} \cdot N + k_{clusters}$ ($k_{clusters} = 0$ when no variant is non-independent). (iv) *Topology-adjusted capture term:* the number of captures is $\max(1, \lceil cycles / k_{cap} \rceil)$, where k_{cap} is the average number of variants reachable per capture. If a shared vulnerability exists (§4.4, mechanisms a–k), the real cost is closer to the homogeneous case.

2.1 Determining which class applies

The dominant class for a platform and program is determined by assessment; the §8 experiment tests the burden under each class and does not determine which applies. The proposed method has four inputs: *capture exposure* (likelihood of physical loss or seizure in the operating area, from operational history and public threat reporting); *extractability* (tamper resistance, storage encryption, key management, and the firmware-extraction effort demonstrated in a red-team capture exercise on the platform class); *remote paths* (remote-intrusion and supply-chain routes to the same artifacts); and *live observability* (whether probing the live platform exposes the state needed to exploit it). The output is a ranked list of acquisition classes per platform, with evidence and the cost ranges assumed for $c_{capture}$, $c_{exploit}$, and c_{public}. A_2-hybrid is treated as dominant when the extraction exercise succeeds at a cost below the value of compromising the fleet; A_2-pure when live observation suffices and extraction fails. Among the RFI's examples, autonomous tactical platforms, robot fleets, and edge networks are exposed to physical capture; secure communication pipelines, industrial control systems, HPC clusters, and cybersecurity applications are more exposed to supply-chain and remote-intrusion A_2-hybrid paths.

3. STATE OF THE ART AND ITS STRUCTURAL FAILURE MODES [RFI 1(iii)]

Each current defense has a specific structural failure mode (L7).

Technique	Structural failure
Redundancy (N-modular, voting)	Correlated failure: identical replicas share failure modes.
Design diversity	Specification correlation: diverse implementations of one specification share its blind spots. Knight and Leveson [R3] measured coincident failures across independently written versions; Littlewood and Miller [R4] model them.
Formal verification	Model–reality mismatch: the proof is of the model, and field failures include what the model omitted (environment, wear, calibration drift).
Red-teaming / penetration testing	Coverage non-transfer: a finding is instance-specific, so cost scales with the number of variants.
Moving target defense	Observable transition: if the adversary can observe the transition, the defense relocates the target. Aimed at A_1-style attacks.

Each technique depends on an assumption about independence or observability: replicas independent, specifications complete, models faithful, coverage transferable, transitions unobserved. For USV/UUV systems, current practice includes encrypted command-and-control links, anti-tamper enclosures, and post-mission forensics. The last two determine whether a capture yields actionable study access and so set $c_{capture}$ for a platform. Supply-chain provenance enforcement and fleet-wide trust-anchor reissuance are two areas where practice is limited.

4. THE ACQUISITION BURDEN [RFI 1(iv)]

4.1 Burden and finding

FLASH states that variation "could reduce the durability and transferability of instance-specific adversary knowledge" (RFI p. 2). The claim holds where the adversary must acquire knowledge of the deployed instance. Heterogeneity raises the acquisition cost against A_1 and A_2-hybrid by the same structural multiplier, at different per-variant unit costs, and leaves it unchanged against A_2-pure. Transfer and acquisition are different quantities; FLASH's thesis is stated in terms of the first and its practical value depends on the second.

The cycle count prices each variant at its cheapest critical path (§6, minimum-path reduction). The formula $1 + H_{d,variant} \cdot (N-1)$ is the continuous form of the count; the discrete count is its ceiling (§4.4).

Defender's burden. N × per-variant verification, plus the cost of computing $H_{d,variant}$. Verification is a property of the artifact; no variant inherits another's assurance. The $H_{d,variant}$ computation is pairwise: the §6 independence test compares each variant-specific node against every other variant, which is $O(N^2)$ in the worst case, and each comparison estimates the adversary effort to transfer an exploit between two variants, a cost of the kind §3 lists for red-teaming. Under A_1 and A_2-hybrid the adversary's burden grows by a factor of N at $H_{d,variant} = 1$ while the defender's grows by N plus the computation; if the computation grows as N^2, the defender's cost grows faster. Both defender costs are incurred once per variant set, and so is the adversary's, because an exploit against an unchanged variant persists. Repeated attacks add to the adversary's cost only where its knowledge is invalidated (a replaced or patched variant, a closed capture channel), and the defender then pays to verify the replacement; the comparison is per variant set, not per attack. Whether heterogeneity pays in absolute terms depends on per-variant exploit cost relative to verification cost. §6 measures the structural quantity ($H_{d,variant}$); §8 measures the behavioral one, adversary acquisition cost C(A, F); their correspondence is the hypothesis under test.

4.2 How heterogeneity mitigates

Vulnerabilities: under A_1 and A_2-hybrid a variant-specific defect is not a fleet defect, provided the variant fraction with independent cheapest paths is high. *Common-mode failures:* diversity attacks correlated failure directly; implementation diversity does not address specification correlation (§3). *Common fault modes:* a fault mode shared across a family is knowledge that transfers, and heterogeneity breaks the transfer where $H_{d,variant} > 0$. *Attack pathways:* supply-chain insertion into a shared component, a published flaw in a common stack, and an exploit developed on one captured platform are reached to the extent the component or stack is varied across the family. A component common to all variants is a hidden centralization (§4.4) and is not reached.

4.3 Dynamic heterogeneity

Under dynamic heterogeneity the acquisition condition becomes a predictability condition. If the sequence follows observable rules (a rotation schedule, a public selection policy), the adversary specializes against it, which is the A_2-hybrid case at fleet scale. If the sequence is randomized over a space the adversary cannot enumerate, the adversary must specialize against the distribution, which is the A_1 case with added cost. Under A_2-hybrid, unpredictability requires that the entropy source and selection policy cannot be recovered from the captured instance; a seed or policy stored on the platform is readable by capture-and-study. Dynamic heterogeneity helps under A_1 and under A_2-hybrid when unpredictable. For A_2-pure the finding covers static and slow-dynamic heterogeneity; dynamics faster than an engagement present a moving target, a different mechanism not analyzed here. The dynamics must be unpredictable to the adversary and verifiable by the defender.

4.4 Toy check and five scenes

Toy fleet. 100 instances, A_2-hybrid. One variant ($H_{d,variant} = 0$): one capture-and-exploit cycle compromises the fleet. N = 10 variants of 10 instances, $H_{d,variant} = 1$: 10 cycles. N = 10, $H_{d,variant} = 0.3$: three variants need independent cycles and seven are compromised from one study, so the exact count is $H_{d,variant} \cdot N + 1 = 4$. That expression holds for $0 < H_{d,variant} < 1$; at the endpoints the exact count is 1 and N. The formula gives $1 + 0.3 \cdot 9 = 3.7$. It equals the exact count at both endpoints, sits below $H_{d,variant} \cdot N + 1$ by $H_{d,variant}$ cycles in between, and its ceiling equals the exact count for every $H_{d,variant}$ in [0, 1]. The check confirms the arithmetic only; whether it holds for a real fleet is the §8 question.

Scene	Fleet (100 doors; ten types in Scene 3)	Result
1	Homogeneous: one lock	1 cycle
2	$H_{d,variant} = 1$: every lock distinct	N cycles
3	$H_{d,variant} = 0.3$: three unique locks, seven types sharing one	4 cycles (formula 3.7)
4	Different hulls, sensors, and vendors; same firmware	$H_{d,variant} = 0$; 1 cycle
5	Shared hallway: one capture reaches k_{cap} variants	captures = $\max(1, \lceil \text{cycles} / k_{cap} \rceil)$; total cost = captures × $c_{capture}$ + cycles × $c_{exploit}$

Hidden centralization (Scene 4). The adversary's knowledge transfers across variants that share: (a) a software stack; (b) a supply chain or single-vendor components; (c) a physical weakness in a material or process; (d) a generator whose low-entropy choices point at the family's failure surface; (e) a specification or protocol; (f) trust anchors, keys, or update channels; (g) a toolchain; (h) an environmental dependency such as GNSS; (i) a deployed perception or classification model, its weights, or its training data; (j) in-envelope behavior, because the invariant set requires equivalent behavior inside the envelope, so an exploit acting through specified behavior (a spoofed contact that induces a compliant maneuver) applies to every variant; (k) sensing phenomenology, such as radar or acoustic spoofing and jamming. The §6 test finds (a)–(c), (f)–(i), and (k) at the node level. Mechanism (d) needs the family-level check in §6. Mechanisms (e) and (j) are specification-level and fall outside $H_{d,variant}$ (L4). Scene 5 is A_2-hybrid; topology changes the capture term, not the class.

What the scenes omit. The defender's burden (§8, criterion 6); real values of $c_{capture}$ and $c_{exploit}$ (§8); independence in a real family, for which published multiversion experiments with recorded coincident failures [R3] are candidates (§6); and the adversary's success criterion. If the goal is less than every variant, effective N is the number the adversary must reach; a family deployed as a single variant has N = 1 for the fielded subset regardless of design-time $H_{d,variant}$.

5. INVARIANTS AND DESIGN DEGREES OF FREEDOM [RFI 1(v)]

The six RFI invariants, specified for the anchor, plus three the RFI does not name:

Invariant	Coverage for the anchor	Status
Functional	Same classification output and avoidance command for the same sensor input	RFI
Safety	Hazard set (collision, grounding, loss of control) unreachable, including degraded-sensor states	RFI
Security	No command injection, loss or diversion, or falsified contact data, within the §2 three-class model	RFI
Timing	Sensor-to-avoidance-command latency and fusion-cycle deadline preserved	RFI
Interoperability	Variants operate within a common fleet; also the channel Scene 5 uses	RFI
Performance	Classification accuracy at declared range and sea state; speed and endurance	RFI
Degradation envelope	Behavior preserved across wear, fatigue, thermal cycling, end-of-life margin	Added
Environmental drift	Behavior preserved across sea state, visibility, weather, temperature	Added
Calibration band	Behavior preserved within field-achievable sensor and actuator tolerance	Added

Functional equivalence is a property of a model, and a platform spends most of its service life away from nominal. An invariant set that omits degradation certifies the system that was tested, not the system that is deployed. COLREGs Rules 6 and 19 name environmental conditions (visibility, sea state, traffic density) in prose; none of the rules addresses wear, fatigue, or thermal cycling.

Degrees of freedom. Implementation (language, scheduling within the timing budget; validity by equivalence checking against the reference model); architectural (sensor suite, redundancy topology, communication graph; requires the invariant set at architectural level); physical (materials, tolerances, thermal path; model must include dynamics, and the degradation envelope becomes load-bearing); algorithmic (estimation, planning, control at equivalent performance; equivalence is behavioral and needs cross-variant testing).

Apparent versus genuine. Two variants are semantically heterogeneous when they differ in which failures are system-failing, and syntactically heterogeneous when they differ in appearance but share the failure structure. The test: vary the DOF, decompose both variants to critical failure nodes, and compare. If the critical node set is unchanged, the DOF is apparent. If it changed, ask whether the exploitation technique developed against variant A gives material advantage (at least θ, §6) against variant B; if yes, the DOF is apparent at the adversary's cost level. Diversity lies in out-of-envelope failure, not in-envelope behavior, which the invariant set fixes; exploits that act through specified in-envelope behavior apply to every variant (L4).

COLREGs illustration. Rule 5 requires “a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions.” The clause “appropriate in the prevailing circumstances and conditions” makes the DOF real: radar, EO/IR, acoustic, LIDAR, and AIS fusion are all legitimate implementations. Two USVs whose different sensor suites degrade in the same sea state under the same failure mode are apparently diverse; an exploit against one applies to both. Rule 17 sets two thresholds of different type: the stand-on vessel *may* act “as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action” (17(a)(ii)) and *shall* act when “collision cannot be avoided by the action of the give-way vessel alone” (17(b)). If either threshold is outside the invariant set, two compliant variants can behave differently in the same encounter. The rules also vary by regime and time: International Rule 28 prescribes lights for vessels constrained by draught, while Inland § 83.28 is [Reserved]; the COLREGs have been amended seven times since 1977. This case applies the framework’s categories to a regulatory specification; it does not show that heterogeneity raises adversary cost for USV/UUV fleets.

6. HETEROGENEITY METRIC [RFI 2(i)]

Surface metrics (variant count, code diversity, failure-mode count) count differences that do not change the adversary’s problem. Published diversity measures quantify failure correlation or code-level variation: Littlewood and Miller [R4] model coincident failure across versions, and Larsen et al. [R5] survey automated software diversity. $H_{d,variant}$ counts the variants whose cheapest exploit path is independent, which is the quantity the §4 formula prices.

Procedure. For a family $F = \{v_1 \dots v_N\}$ generated from one specification, decompose each variant to its critical failure nodes $N_c(v, d)$: components whose failure produces system-level failure within the declared model scope, weighted by consequence, at declared depth d. The criticality threshold is declared.

Step 1: independence test. A node X in variant v that appears in no other variant is *variant-specific* if no technique developed against X gives material advantage against any node in any other variant, and *common* otherwise. Material advantage is a reduction in adversary cost of at least a declared threshold θ (for example 20%); a transfer below θ counts as none. The test is binary because $H_{d,variant}$ requires a binary classification, and it operates on (node, variant) pairs for partially shared nodes. S_c is the set of variant-specific nodes.

Step 2: node ratio (diagnostic). $H_{d,node} = |S_c| / |\cup_v N_c(v, d)|$.

Step 3: minimum-path reduction. For each variant, find the cheapest critical path, measured in expected adversary effort in hours for a declared capability class. Effort is per-node exploit development less the transfer discount from the study of any single other variant (the largest discount applies; discounts from studying several variants are not cumulated). Paths within 20% are co-cheapest; the variant is non-independent if either is common and independent if both are variant-specific.

Step 4: variant fraction. $H_{d,variant} = |\{v \in F : \text{cheapest critical path of v is variant-specific}\}| / |F|$. This is the quantity in the §4 formula.

Family-level generator check. If the family's designs come from a process whose biases are inferable from a small sample, the family is non-independent and $H_{d,variant}$ is reported as 0. This is a human judgment (L5).

Analyst-judgment points. *J-1, node granularity:* the default node is a specific artifact (part number and firmware version); treating a functional class ("radar") as one node is a permitted alternative that classifies more nodes as common and lowers $H_{d,variant}$. *J-2, generic versus specific transfer:* a technique any adversary could develop without capturing a variant is background knowledge and does not count as transfer. *J-3, cheapest path in what unit:* as in Step 3. The report states the choices made.

Worked example separating the two quantities. N = 10 variants, each with 1 shared and 9 variant-specific critical nodes. The union has 91 nodes and S_c has 90, so $H_{d,node} \approx 0.99$. If the shared node is cheapest for every variant, one exploit compromises all ten: $H_{d,variant} = 0$. The formula with $H_{d,node}$ predicts $1 + 0.99 \cdot 9 \approx 9.9$ cycles; the correct prediction is $1 + 0 \cdot 9 = 1$. The formula uses $H_{d,variant}$. The two diverge in the other direction when a variant's cheapest path is variant-specific but most of its nodes are shared.

Direction of error. $H_{d,variant}$ approximates $H_{variant}$, the fraction under the true critical failure set, and its error runs both ways. Two conventions lower it: nodes below depth d are counted as common, and the co-cheapest rule classifies a variant as non-independent if either path is common. Three can raise it: the J-1 default granularity, the binary threshold θ with discounts below θ not cumulated, and a false no-transfer call in Step 1. The report lists the conventions used.

Bridge and independent check. FMEA failure nodes serve as a proxy for exploitable surfaces; the proxy is asserted, not proven, and its validity is a per-domain question that §8 tests. To check $H_{d,variant}$ independently of §8, compute it on families with documented cross-variant failure history. Where failures recurred across variants, $H_{d,variant}$ should be low; where they were variant-specific, high. This tests the recurrence correspondence only. Where the domain lacks such history, the check can run on published multiversion experiments [R3] or on synthetic families with injected, documented correlations.

7. VALIDATION OF EQUIVALENCE AND INVARIANTS [RFI 2(i); thrust (ii)]

A proof establishes a property of a model. Three model scopes need different evidence. *I/O equivalence:* differential fuzzing, test-vector conformance, formal equivalence checking against the reference model. *Plus control and timing:* timed-automaton equivalence, statistical model checking, hardware-in-the-loop testing. *Plus physical dynamics:* co-simulation with validated plant models, residual estimation against field data, envelope testing at declared boundaries. Equivalence at the first scope is achievable with current formal methods; the assurance problem lives at the third, where the degradation envelope, environmental drift, and calibration band enter as required invariants. The claim for each variant is equivalence within the stated model scope, with model–reality mismatch carried as a declared residual. Other invariants have their own validation: safety by reachability analysis over a hazard set that includes degraded states; security by the §2 three-class model; interoperability is the one invariant whose statement demands cross-variant evidence.

8. EVALUATION CRITERIA AND PROTOCOL [RFI 2(i), 1(iv)]

Six criteria. A heterogeneity claim is assessable if and only if it states: (1) the variant family and its critical node set, with J-1 to J-3 applied and the independence test and minimum-path reduction run; (2) the measured $H_{d,variant}$, the depth, and the analyst's range; (3) the adversary class, with the observation and capture model specified; (4) time-to-compromise T_f under a calibrated autonomous exploit-development agent, with the fraction f declared (f = 1 for full-fleet compromise; for f < 1 the effective N is f·N); (5) the correspondence between $H_{d,variant}$ and T_f, as a test with interpretation fixed in advance; and (6) the defender's cost: per-variant verification effort and the effort of computing $H_{d,variant}$, recorded in the same experiment.

Protocol. *Calibration:* run the agent, capable of adaptive search, hypothesis revision, and independent operation, against a named known-vulnerable single-variant baseline; at least five runs must each reach a working exploit within budget T_{cal}, or the run is void. Calibration shows the agent can find a known exploit; T is reported as agent time and mapped to adversary cost only under the declared capability class. *Baseline:* a single-variant family, T_{f1}. *Heterogeneous families:* sweep $H_{d,variant}$ from about 0 to high and record T_f for f·N variants. *Three adversary conditions:* A_1, A_2-hybrid, A_2-pure, with the channel declared per experiment. *Capture cost:* T measures exploit-development time after the declared channel is granted; $c_{capture}$ is set by the channel declaration and estimated separately (§2.1), so T alone does not measure $c_{capture} + c_{exploit}$. *Defender cost:* verification effort and $H_{d,variant}$ computation effort are recorded for each family.

	A_1	A_2-hybrid	A_2-pure
Sees	Specification, public documentation, one reference variant	Deployed instance through a declared capture channel, with time to study	Deployed instance through live observation, without study
Channel	None into the deployed instance	Named: capture duration, firmware extraction path, intrusion path	Named: live probing, side-channel access, no persistent access
Topology	n/a	k_{cap} declared; sets the number of captures	n/a

Statistical specification. At least five runs per ($H_{d,variant}$, class) cell; fewer is reported as preliminary. The prediction is multiplicative: for f = 1 the predicted ratio $T_f(H_{d,variant}) / T_f(0)$ is $\lceil 1 + H_{d,variant} \cdot (N-1) \rceil$. Support requires that the 95% confidence interval of the observed ratio excludes 1 and lies within a declared tolerance of the prediction (a design choice, for example a factor of 2); the interval comes from the runs in each cell. Reports state whether mean, median, or geometric mean is quoted (search times are right-skewed), name the baseline, and give the analyst's $H_{d,variant}$ range with the formula value at both ends.

Observation	Interpretation (fixed before the run)
T rises with $H_{d,variant}$ under A_1 and A_2-hybrid; flat under A_2-pure	Supported for both classes; the multiplier is shared
T rises under A_2-hybrid; flat under A_2-pure (A_1 not run)	Supported for A_2-hybrid
T flat under A_1; rises under A_2-hybrid	Structural-symmetry claim falsified
T rises under A_1; flat under A_2-hybrid	Falsified for A_2-hybrid
T flat under all three	Falsified; $H_{d,variant}$ is not predictive
T rises under all three	Falsified; heterogeneity imposes cost under A_2-pure, which §4 did not predict
T flat under A_2-hybrid; rises under A_2-pure	Falsified in the opposite direction
T_{cal} not met	Run void

Formal versus empirical. §8 is an empirical protocol. The formal component is §6's $H_{d,variant}$ procedure, a structured computation with three analyst-judgment points, not a proof. It makes heterogeneity measurable by a stated procedure, the precondition for assessing FLASH's central thesis; the metric's correspondence to adversary cost is tested empirically.

9. SYNTHESIS AND ARCHITECTURE [RFI 2(ii), 2(iii); thrust (iii)]

Placement of the symbolic layer. *Generator:* the symbolic layer produces the design and the model elaborates it; strongest formal guarantee, narrowest generative range. *Checker:* the model proposes and the symbolic layer verifies; broadest range, with verification cost scaling with n. *Constrainer:* the model generates within a symbolic envelope; the envelope is proven and the interior is not. The choice decides whether thrust (ii)'s guarantee is available per variant at generation time or only afterward. The generator is also a potential hidden centralization (§4.4, mechanism d): consistent choices across a family give an adversary who learns the bias a discovery advantage without a capture.

Structurally constrained synthesis. The constraint makes the generator assurable: an unconstrained generator can propose designs the verifier cannot check at acceptable cost. A structurally constrained generator proposes only designs within the reference model's scope. Oracle-guided inductive synthesis imposes the constraint by rejecting candidates outside scope; combining a generative multimodal model with an interactive theorem prover makes the prover's rejection the constraint and its acceptance the assurance.

Single-agent and multi-agent architectures. Single-agent synthesis centralizes the design loop and makes provenance traceable; its failure is a single point of generator bias. Multi-agent synthesis divides proposal, critique, and verification; its failure is shared training lineage, which

shares the bias the separation was meant to remove. What matters for assurance is whether the verifying agent is independent of the generating agent.

Trust transfer and independence. If the same model class generates a design and its proof, the proof states what that model believes. Data independence (different training corpora) gives weak protection, since the same architecture shares inductive biases; architectural independence (a language model generates, a theorem prover checks) is the minimum the requirement needs; process independence (different organizations or teams) is strongest and not always feasible. If the specification is itself model-generated, mis-formalization is the failure mode, so the full requirement is architectural independence between design and assurance plus independent validation of the specification, by a domain expert, a reference model not generated by the same process, or a formal property check against an established standard.

Explanation. An assurance artifact establishes that a property holds; an explanation artifact says why, in terms a reader can follow. Explanation fails by persuasiveness without faithfulness, and its comprehensibility correlates with persuasiveness. The criterion: the explanation is architecturally independent of synthesis and is checked against the assurance artifact by a named checker who is neither the explanation's author nor the synthesis process. The checker uses derivation (derive the proof's claim from the explanation) or counterexample search (construct a system that satisfies the explanation but not the proof, or the reverse). A model used as checker must be architecturally independent of the explanation generator; a human checker's method and basis are recorded. Understanding is the channel through which the assurance residual reaches deployment.

10. REUSABLE FORMALIZATION LIBRARIES [RFI 2(iv)]

Reusability across variants depends on the level of abstraction relative to the §5 degrees of freedom. Reusable: the specification of required behavior, a reference model at a chosen scope with its proof, and a failure-history corpus (the input to §6's decomposition). Partly reusable: proof tactics and lemma libraries, and specification-level tests. Not reusable: per-variant proofs and instance tests, because a library that stores them has stored the variants. The reference model's scope is versioned, because the scope changes when the invariant set is extended (§5); the COLREGs have been amended seven times since 1977, and a library without scope versioning conceals which version its members are proven against.

11. ASSUMPTIONS, LIMITATIONS, AND FALSIFIERS [RFI 1(iv)]

Assumptions.

- **A1 Acquisition condition:** heterogeneity raises adversary cost under A_1 and A_2-hybrid by the multiplier $1 + H_{d,\text{variant}} \cdot (N-1)$ and does not affect it under A_2-pure for static or slow-dynamic heterogeneity; the class applying to a program is determined per §2.1.
- **A2 Model scope:** variants are generated under one specification whose scope is declared.
- **A3 Measurability:** the §6 decomposition applies to the class, and the FMEA-to-adversary-cost bridge is asserted, not proven; the formula has no usable input if A3 fails, and §8 is the test.

- **A4 No missed transfer.**
- **A5 Post-test $H_{d,variant}$:** a formula evaluated with $H_{d,node}$ overstates the burden when a cheap shared path is invisible in the node ratio and understates it in the opposite case.
- **A6 Topology-adjusted capture term.**
- **A7 Cluster structure:** a single transfer cluster, or $k_{clusters}$ reported beside $H_{d,variant}$.

Limitations.

- **L1** The correspondence between $H_{d,variant}$ and adversary cost is a hypothesis (A3); §6's independent check tests recurrence and §8 tests adversary cost.
- **L2** $H_{d,variant}$ approximates $H_{variant}$ with error in both directions (§6).
- **L3** The anchor is named from public regulatory documents. The response claims no program participation, fleet data, or operational experience in the domain. The USV/UUV domain may lack the documented cross-variant failure history §6's independent check requires, and item RFI 1(i) is addressed at the class level with no platform or program proposed.
- **L4** Exploitation through a shared specification or through specified in-envelope behavior (§4.4, mechanisms e and j) applies to every variant by construction and is not priced by the formula; under it the burden is the homogeneous case regardless of $H_{d,variant}$.
- **L5** $H_{d,variant}$ depends on the analyst at J-1 to J-3: the analyst estimates adversary effort in hours and judges whether a transfer meets θ, and two analysts may differ. The report gives the value used and the range, with the formula value at both ends.
- **L6** The defender's burden includes an $H_{d,variant}$ computation that is potentially $O(N^2)$.
- **L7** §3 is a reading of the field and position, not measurement; no systematic literature survey was conducted.
- **L8** A2, A6, and A7 bound where the formula applies; outside them its cycle count is not a prediction.

What would falsify the argument.

- Flat time-to-compromise under A_2-hybrid (§8): $H_{d,variant}$ would not predict adversary cost where the argument claims it pays.
- $H_{d,variant}$ uncorrelated with documented failure history (§6): the metric would not measure what §6 claims.
- T rising under A_2-hybrid but flat under A_1 (§8): the structural-symmetry claim in §4.1 would be wrong.

Program-feasibility conditions.

- If no architecturally independent assurance path can be built within program constraints, the independence requirement is not a program option.
- If $H_{d,variant}$ cannot be computed at a useful depth for real systems, the metric is not usable.
- If the specification-formalization path cannot be implemented, the full independence requirement cannot be met.

Outcomes that change the practical claim.

- A_1 dominance in every RFI example keeps the structural multiplier at a lower absolute per-variant cost.
- A_2-pure dominance in the RFI's examples removes the benefit of static heterogeneity, because that class is unaffected by it.

Next step.

- A pilot on one variant family in a domain with public cross-variant failure history.
- The pilot first verifies that the domain has such history; computes $H_{d,variant}$ with J-1 to J-3 applied and the range declared; records the deployment topology and k_{cap}; applies §2.1 to determine the dominant class; runs the §8 protocol under all three classes; and tests the correspondence between $H_{d,variant}$ and T_f.
- Two analysts independently apply §6 to the same variant pair; the divergence in their $H_{d,variant}$ values is the measured analyst-dependence residual.
- The pilot reports a validated or a falsified metric.

ANNEX A. RFI COVERAGE MAP

RFI ask	Section
1(i) Operational domain and target application	§1
1(ii) National-security rationale and threat environment	§2, §2.1
1(iii) State of the art; limitations; points of failure	§3
1(iv) Mitigation; assumptions; limitations; evaluation methods	§4, §8, §11
1(v) Invariants and degrees of freedom	§5
2(i) Heterogeneity metrics; equivalence validation; adversarial experiments	§6, §7, §8
2(ii) Structurally constrained induction and deduction	§9
2(iii) Single-agent and multi-agent architectures	§9
2(iv) Reusable libraries	§10

Background claims and thrusts: symbolic reasoning with generative agents, §9; decoupling requirements from instance-level design, §5 and §7; meaningful variation with preserved properties, §4 and §5; reduced durability and transferability of adversary knowledge, §4; common fault modes, §4.2; semantic heterogeneity, §5; formal versus empirical assessment, §8; dynamic heterogeneity, §4.3; thrust (i), §5; thrust (ii), §7; thrust (iii), §9 and §10.

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COLREGs citations are from the IMO Convention text (1972, as amended through the 2013 amendment in force 1 January 2016); 33 CFR Part 83 is cited as published in the eCFR on 5 November 2026.

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